

Ronald Luo

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Software Engineer & Applied ML Researcher

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EDUCATION

- **McMaster University** Hamilton, ON
Master of Science in Biomedical Engineering (\$28k research funding, 3 publications pending) *Aug 2021 - April 2025*
- **McMaster University** Hamilton, ON
Honours Bachelor of Science in Life Sciences *Aug 2016 - Jul 2020*

WORK EXPERIENCE

- **MedBrewery (EdTech Startup)** Toronto, ON
Full-Stack Engineer *Jan 2025 – Present*
 - **Platform:** Built MCAT prep platform (Next.js, TRPC, Prisma, Tailwind, Stripe) with daily questions, exam bundles, tiered pricing, and analytics (Google Sheets + webhooks).
 - **Organic Growth:** Drove 150+ signups by building Reddit, campus, Discord, and subreddit campaigns along with dynamic Open Graph previews, as measured by Google Analytics traffic flow and UTM parameters.
 - **AI/Automation:** Developed multimodal agentic tool for personalized content generation and adaptive user guidance, a core feature of the platform.
- **ESPN Bet / theScore Bet (Penn Entertainment)** Toronto, ON
Software Development Intern *May 2023 – Dec 2023*
 - **QA Automation:** Built Playwright test suites for Barstool to ESPN Bet migration, saving 300+ QA hours.
 - **Training:** Led DataDog logging workshop for 50+ stakeholders, improving organization monitoring practices.
 - **Regulatory Compliance:** Managed functional testing and sign-off for staging and UAT test environments, ensuring iGaming regulatory compliance.
- **Delta Hacks (Major League Hacking)** Hamilton, ON
Software Engineer *Sep 2022 – Sep 2023*
 - **Website Redesign:** Updated website branding with design team, resulting in a 22% increase in user traffic (Google Analytics) compared to previous year.
 - **API Optimization:** Migrated from REST to tRPC and implemented caching, making repeat requests ~65% faster than cold requests (Vercel logs).
- **Zafin (Digital Banking Platform)** Toronto, ON
Software Engineer Intern *May 2022 – Aug 2022*
 - **Analytics:** Built ReactJS Redux web app for Canadian banks, improving analytics from quarterly to real-time.
 - **Agile Delivery:** Contributed to sprint planning and feature delivery with cross-functional teams in fintech.

ACHIEVEMENTS

- **Cecil & Yvette Yip Graduate Award:** For innovative interdisciplinary research in biomedical engineering (2023)
- **Top 50 Replit Creator Award:** Replit Top 50 Creator for Collaborative Sketchbook project (2022)

PROJECTS

- **100-Websites Portfolio:** Developed and deployed web apps exploring frameworks, design patterns, and deployments.
- **AI Literature Screening Tool:** Built with LangChain, vector storage, and RAG to automate literature screening.
- **AI for Literature Screening Preprint:** First-author medRxiv study benchmarking LLMs as research tool.
- **IMU Sensor Data App:** Smartphone app for IMU data collection and gait classification with deep learning.

PROGRAMMING SKILLS

- **Languages:** Typescript/JavaScript, Python, Java, C++, C#, SQL/NoSQL, HTML, CSS
- **Technologies:** React, Next.js, Node, Express, Git, Redux, PostgreSQL, MongoDB, Docker, Playwright, Jest
- **Machine Learning:** Classical ML, Deep Learning (RNNs, CNNs), Transformers, PyTorch, Keras, TensorFlow, Scikit-learn, Hugging Face, LangChain.js, Vector Storage (FAISS, HNSW)